



Population Challenges

Implications of population dynamics to the provision
of informal care: a comparative analysis between
Spain and the United Kingdom

Thiago Cordeiro Almeida

Recent population changes - challenges and opportunities?

Recent demographic trends have highlighted the challenges of the ageing population to economic and social sustainability (Goodhart and Pradhan 2020).

- On one hand, the share of workers has declined, while the consumption behavior of the older population has changed (Mason and Lee 2022).
- On the other hand, the decrease of the young population (in absolute and relative terms) offers an opportunity to improve the quality of their education and diminish the burden of housework attributed mainly to women.

In this sense, depending on which **social policies governments adopt** and the **demographic changes** observed in each population, the **burden of care under the responsibility of the families** can vary.

Motivation

Alternative arrangements of social care systems have been debated, mainly due to the recent ageing of the population observed in European countries and worldwide.

- In Europe, cultural differences among regions have resulted in different types of familistic structures of the welfare system (Reher 1994, Zagheni et al. 2015, Van Eck et al. 2024).
- **Comparison between Spain and the UK:** it can reveal how different contexts with different structures of policies could be affected by demographic changes (Egea 2023; Curry 2023).

In this debate, going beyond the household structure and considering all the **kinship networks** can provide a fundamental demographic perspective to understand these potential impacts on society.

Research Question

What are the differences in the potential availability and composition of kinship to provide informal care for the population between Spain and the United Kingdom?

- Are there differences regarding the potential availability of care providers in the kinship network?
- Are there differences regarding the potential composition by gender and type of relative to provide care?
- What are the differences between the results obtained and using a “simpler” approach, considering “usual dependency ages”?

The availability and composition of kinship to provide care

Kinship networks are essential to social support, such as informal care, wealth transfers, and emotional support (Alburez-Gutierrez et al. 2023).

- Elderly relatives play a role not only as receivers of informal care, but also as producers for younger generations (Adhikari & Alburez-Gutierrez 2025; Margolis and Verdeny, 2019)
- Even though some relatives live in other households, some types of relatives are more likely to provide social support (Hunteler et al., 2025) or would be potentially demanded to support other relatives (Alburez-Gutierrez et al. 2021).
- The 65+ yo population living in countries strongly affected by outmigration would potentially have a lower availability of relatives for informal care (Matos-Moreno et al. 2025)
- Subnational (Hunteler et al. 2024) and socioeconomic (Song and Mare 2019) heterogeneities can play a role in this debate

Comparison between countries with different care policies and considering a refined measure to capture “informal care provision” is less present in this debate.

Definitions

Provision of informal care

- Care
 - Economic sector, contributing significantly to the GDP (Guimarães and Pinheiros 2023).
 - It can be based on paid or unpaid work
- Informal care
 - It is considered all the activities directly or indirectly related to care (Guimarães and Pinheiros 2023), realized outside the formal labor market

Data

Aggregate data:

- **Fertility:** Age-specific-fertility rates (ASFR) from World Population Prospects (WPP), UN/DESA, Review 2024.
- **Mortality:** Life table by sex from WPP, UN/DESA, Review 2024.

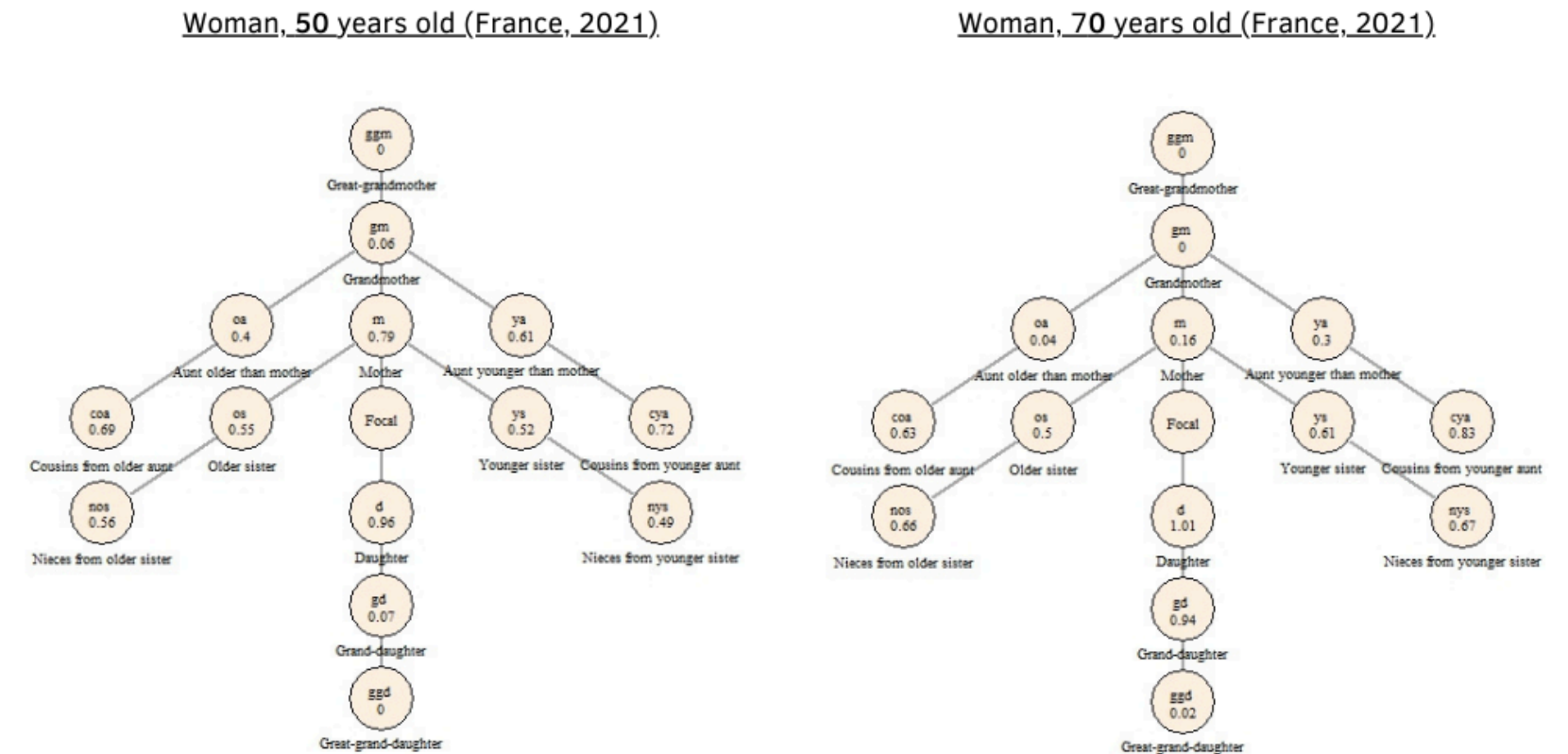
Time use data:

- **Production and consumption:** Age-specific measures of production and consumption of unpaid work from National Time Transfer Account (NTTA).

Method (1)

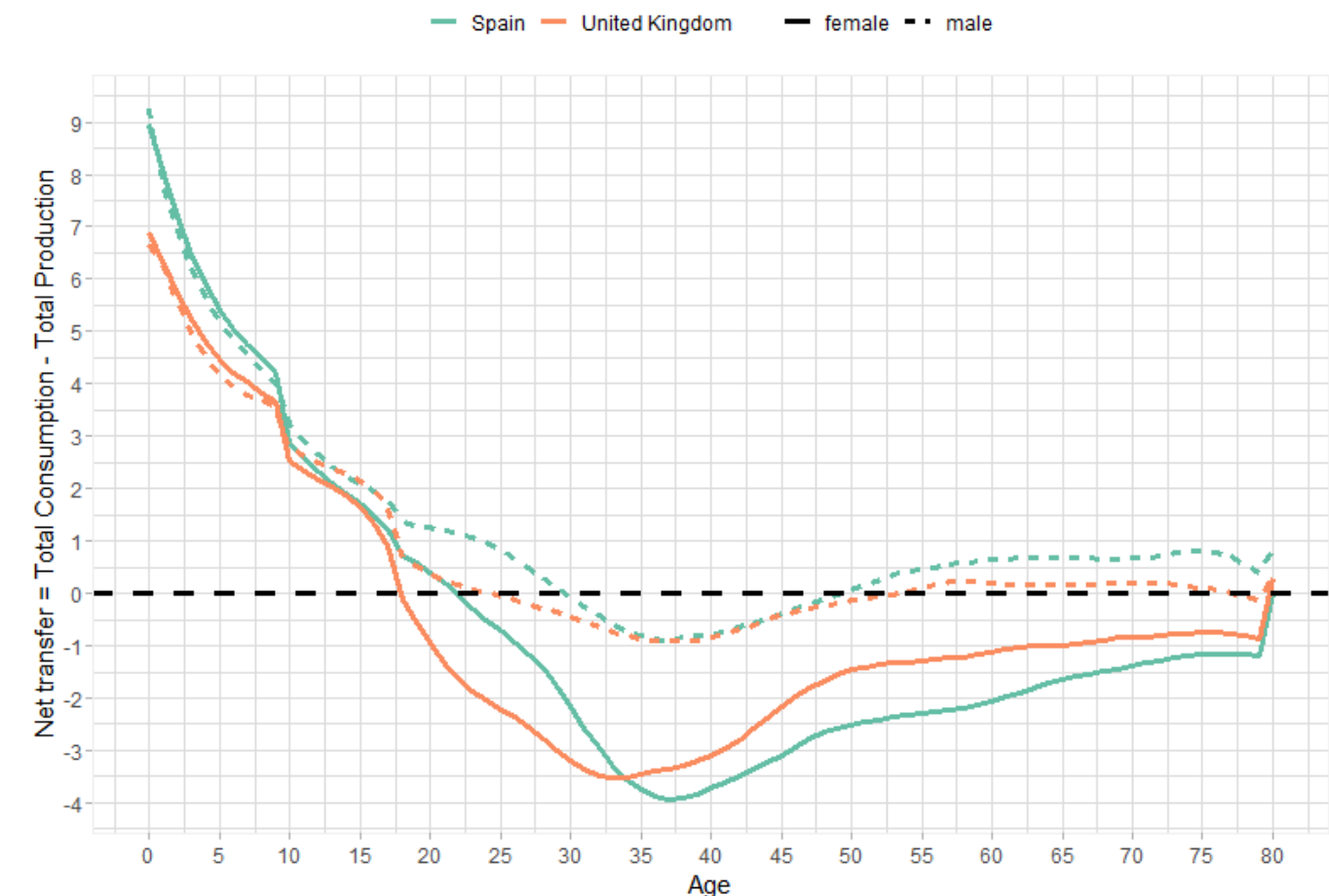
Demographic availability of kin (A):

- Time-varying two-sex model based on matrix relations (Alburez-Gutierrez et al. 2023).



Time transfer (B):

- Age-specific average hours of time transfer by sex.

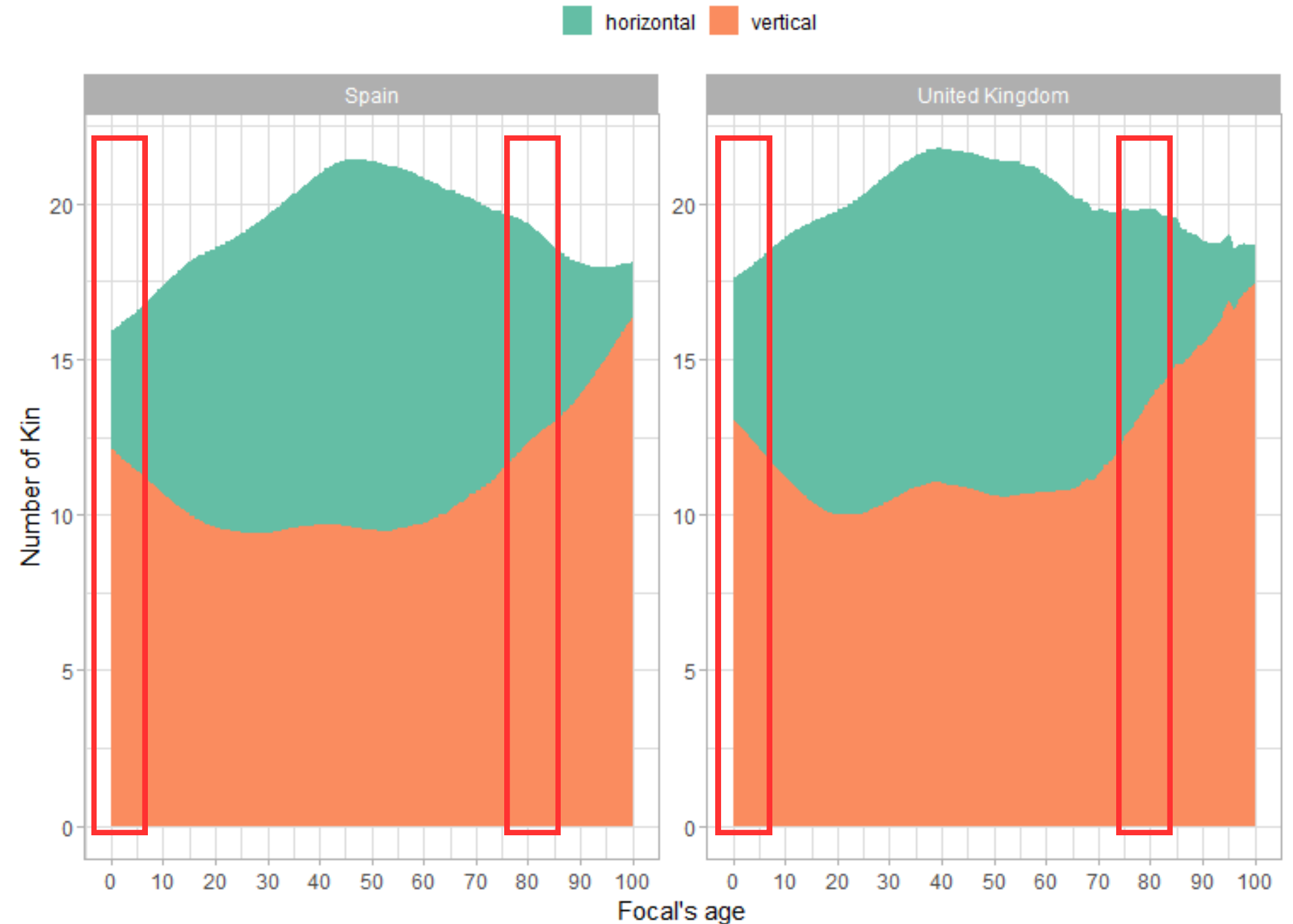


Source: NTTA.

Method (2)

Combining both (A + B):

- From A:
Compute the number of relatives by sex and type of relative for a focal aged 0 and 80.
- From B:
Take the age limits to be considered a net producer of unpaid work for each country.



Source: UNDESA, World Population Prospects, Revision 2024.

Measure

Kinship Dependency Ratio (KDR):

After computing the number of relatives for each focal person aged 0 and 80, we have to consider that these relatives can also have other dependents to take after.

To equalize that, we can calculate the KDR:

- **Numerator:** number of kinship that potentially can provide informal care
- **Denominator:** number of kinship that potentially can demand informal care.

Interpretation: for a certain age of a focal person, they can have a potential number of providers of informal care, on average.

Expected results

Are there differences regarding the potential availability of care providers in the kinship network?

- Yes. A higher availability is expected in the UK due to demographic changes (higher fertility) and a lower age at entering the net provider age group.

Are there differences regarding the potential composition by gender and type of relative to provide care?

- Maybe. Once both countries have different demographic behaviors, it can create a different number of relatives

What are the differences between the results obtained and using a “simpler” approach, considering “usual dependency ages”?

- It is expected to find higher differences in Spain compared to the UK due to the different ages of entering the net provider age group.



Population Challenges

Thank you!
Merci!

Thiago Cordeiro Almeida
thiagocordalmeida@gmail.com

Some references

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